

Technique 3: Hollow Objects

MP

Teach You Physics Hand In Hand

April 2026

1 SOLUTION to Problem in Approximation Post

- Level of difficulty: **5/10**

a) Summing gravitational forces, x -components cancel, y -components add:

$$F_y = -\frac{2GMm y}{(a^2 + y^2)^{3/2}} \quad \text{4 points}$$

For small $y \ll a$, $(a^2 + y^2)^{3/2} \approx a^3$:

$$F_y \approx -\frac{2GMm}{a^3} y \quad \text{3 points}$$

This is simple harmonic motion with $k = \frac{2GMm}{a^3}$, so period:

$$T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{a^3}{2GM}} \quad \text{1 points}$$

bi) For small $d\Omega$, the area is simply a small circle. Using small angle approximation:
Area:

$$dA = \pi R^2 \sin^2 \frac{d\Omega}{2} \Rightarrow \frac{1}{4} \pi R^2 d\Omega^2 \quad \text{1 point}$$

Circumference:

$$2\pi R \sin \frac{d\Omega}{2} \Rightarrow \pi R d\Omega \quad \text{1 point}$$

bii) The gravitational field on the y -axis from two masses M at $(\pm a, 0)$ is

$$g_y(y) = -\frac{2GM y}{(a^2 + y^2)^{3/2}}$$

For small displacement $\vec{\delta} = (\delta_x, \delta_y, \delta_z)$ from the centre at $y = A$,

$$dF_y = dm \delta_y \left. \frac{dg_y}{dy} \right|_{y=A} \quad \text{1 point}$$

The derivative:

$$\frac{dg_y}{dy} = -2GM \frac{a^2 - 2y^2}{(a^2 + y^2)^{5/2}}$$

At $y = A \ll a$, binomial approximation gives:

$$\left. \frac{dg_y}{dy} \right|_{y=A} \approx -\frac{2GM}{a^3} \quad \text{3 points}$$

Integrate over the hemisphere ($0 \leq \theta \leq \pi/2$):

$$F_{\text{half}} = \rho \left(-\frac{2GM}{a^3} \right) \int_0^r \int_0^{\pi/2} \int_0^{2\pi} (r' \cos \theta) r'^2 \sin \theta d\phi d\theta dr' \quad \text{2 points}$$

$$F_{\text{half}} = -\frac{\pi GM \rho r^4}{2a^3}. \quad \text{2 points}$$

Tidal stress (force per unit area on equatorial plane πr^2):

$$\sigma = \frac{|F_{\text{half}}|}{\pi r^2} = \frac{GM \rho r^2}{2a^3} \quad \text{2 points}$$

2 Finding a field in a hollow part

When we encounter question involving **asymmetric** composition of mass or charge with a hollow part in an object, of course we can solve Maxwell's set of PDEs but that would be too annoying. Therefore, here we introduce a useful technique: **negative quantity**, which is separate the original object and the hollow part as two distinct objects and set the mass or charge of the hollow object to be some negative values. And we can find the field or anything inside the hollow part.

3 Example

Question: Consider a sphere with evenly-distributed charge with charge density ρ . The radius of the sphere is R . Now, some part inside the sphere is removed, so that a small hollow sphere of radius r is created inside the sphere. Distance between centers of the two spheres is a . Find the electric field distribution inside the hollow part.

Answer: Assume the hollow sphere has charge density of $-\rho$. Separate the original asymmetric sphere into a sphere without hollow part with charge density ρ and radius R and a small sphere with charge density $-\rho$ and radius r . When superimposed together, they result in no charge in hollow part. Consider sphere without hollow part. By Gauss Law:

$$\oiint E ds = \frac{Q}{\epsilon_0} = \frac{\rho \frac{4}{3} \pi a^3}{\epsilon_0}$$

$$\vec{E} = \frac{\rho \frac{4}{3} \pi a^3}{\epsilon_0 4 \pi a^2} \hat{a} = \frac{\rho a}{3 \epsilon_0} \hat{a}$$

Electric field contributed by the small sphere is 0 at its center, so electric field at the center of small hollow sphere is simply:

$$\vec{E} = \frac{\rho a}{3\epsilon_0} \hat{a}$$

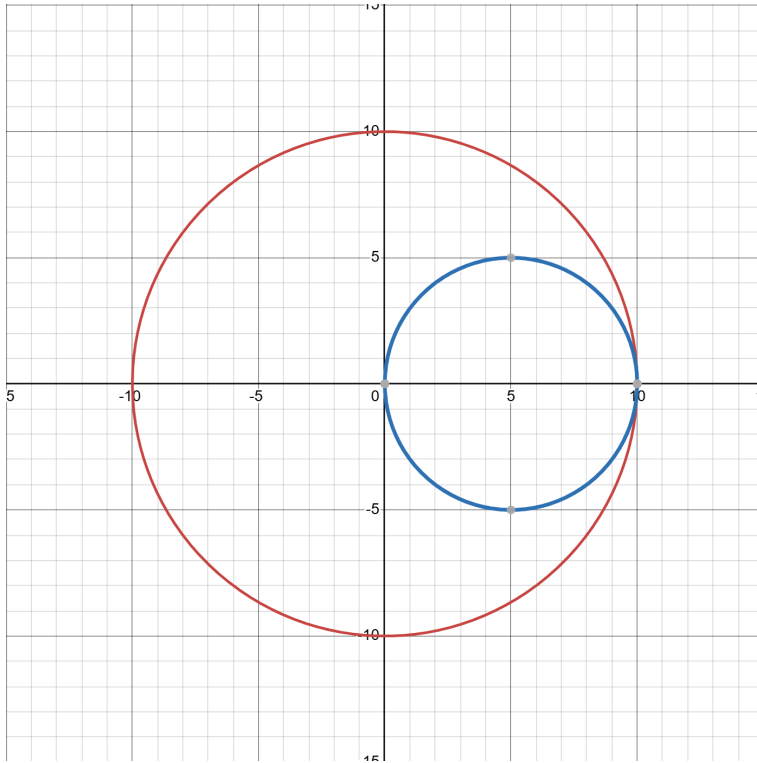
Since we have to find the electric field distribution in the whole hollow part, we can find the electric field at random distance $\vec{b}$ away from the center of the hollow sphere. Define $\vec{c} = \vec{a} + \vec{b}$, where $\vec{c}$ is displacement vector from center of large sphere to the random point. We have:

$$\vec{E} = \frac{\rho c}{3\epsilon_0} \hat{c} + \frac{(-\rho)b}{3\epsilon_0} \hat{b} = \frac{\rho a}{3\epsilon_0} \hat{a}$$

The electric field distribution is uniform inside the hollow sphere and the problem is solved.

4 Problem

Consider a thin circular disc of radius R . The mass of the disc is M . A hollow part in circular shape with radius $\frac{R}{2}$ is introduced in the disc. The distance between the centers is $\frac{R}{2}$ so that the center of the disc lies on the circumference of the hollow circle. Find the position of center of mass of the disc.



The solution will be uploaded in the next post and MP love you all :)